a
False-positive claims (%)
54.5%
0.0%
Ungated (n = 55)
Gated (n = 20)
True-positive claims
25
20
b
Ungated
Gated
Trusted-deterministic
Untrusted-oracle
GC target
U reduction
Purine band
Angiopoietin-2
IL-17A
PDGF-BB
SARS-CoV-2 N
Thrombin
VEGF
FGF2
IFN-γ
swine H3 HA
True positive
False claim
Suppressed
Borderline
c
59 → 0
Overreach events
Ranking
Screening
RhoFold
ProteinX
EGARPS
Ungated
Gated
d
Ungated
Gated
CAI report
A1AT
Insulin
EPO
GFP
MFE report
Constraints
Relative comparison
3′UTR stability rank
Cross-family MRL
Novel 5′UTR 7
Novel 5′UTR 11
Novel 5′UTR 13
Junction quality
MRL optimum
EGFP random-50
Var-len human
True positive
False claim
Neutral
No published claim
e
allow · pass n = 4
downgrade n = 2
blocked n = 3
DecisionGate (pre-action)
5′UTR MRL
3′UTR
certified · EGFP random-50
certified · var-len random
certified · var-len human
never-profiled family
certified family
executed · MRL 6.28 = Exp-14
not blocked
executed · MRL 6.32
blocked pre-call
ClaimGate (pre-publication)
5′UTR claim · certified
5′UTR claim · never-profiled
3′UTR claim · never-profiled
absolute scope stands
scope → within-target relative
no publishable basis
allow / pass
downgrade
require-profiling
reject
f
RhoFold pLDDT
CRGPT handle
RNASTOP stab.
GEMORNA 3′UTR stab.
GEMORNA 5′UTR MRL
ProteinX ipTM
EGARPS native prob.
CRISOT off-target
IRESfinder
GerNA-Bind
RNAsmol
reject
require profiling
downgrade
Gate decisions (count)
h
Ungated (Exp 5)
Gated (Exp 17)
mRNA dual-arm (Exp 18)
Decision matrix (Exp 23)
Production (Exp 24)
Review off (Exp 25)
Review on (Exp 25)
Review outage (Exp 25)
45 tasks · 2.0 calls/task
40 runs · 2.0 calls/run
12 cells · 0.0 calls/cell
241 runs · 2.0 calls/run
31 sessions · 0.0 calls/session
31 sessions · 4.6 calls/session
31 sessions · 0.0 calls/session · 144 failed
Task index
LLM calls
g
Review off
30/31 succeed · 0 calls
Review on
30/31 succeed · 144 calls
Review on, API outage
30/31 succeed · 144 failed
mRNA CDS
Full-length mRNA
siRNA
sgRNA
Secondary structure
Tertiary structure
Aptamer
IRES prediction
mRNA scoring
RNA–small molecule
ASO screening
Succeeds · review completed
Succeeds · no review
Succeeds · review unavailable
Failed (PA_023, all arms)